Universal Time-Consistent (TiC) Credit Rating

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Abstract

This paper, based on the author's extensive practical experience and theoretical research, addresses several issues in credit rating that became evident during the Global Financial Crisis of 2008. In particular, it responds to a proposal made by the U.S. Congress and the U.S. Securities and Exchange Commission in a 2012 report advocating the unification of credit rating systems worldwide. The paper proposes an innovative credit rating approach and algorithm - the Time-Consistent Rating Method. Unlike the statistical approaches commonly used in existing credit rating methodologies, this framework employs a unified mathematical formula to generate ratings across different rating systems (including those of banks, professional rating agencies, and academic models), different time perspectives (Through-The-Cycle and Point-In-Time), and different types of rating entities (such as public corporations, and small and medium-sized enterprises).The proposed framework addresses several key limitations of existing credit rating systems. These include the instability and inaccuracy of converting ratings across different systems and methodologies, the significant variation in empirical default rates within the same rating category, and the lack of continuous rating measures in traditional agency ratings. In addition, the Time-Consistent rating framework offers two distinctive advantages. First, it provides a practical solution for rating large corporations or credit exposures where historical default data are limited or unavailable. Second, it allows regulatory credit requirements - such as capital adequacy and loan loss reserve requirements - to be quantified and directly linked to credit ratings. One application presented is a No-Regulatory-Arbitrage rating conversion between different rating systems.

1 Background and Credit Rating 1.0

Credit rating serves as a cornerstone for banks' credit risk management. Important decisions are often made based on customers' ratings. The Basel Accord uses credit ratings as building blocks for risk measuring, monitoring, controlling, and reporting. Basel also standardized credit risk management languages and terms such as "Credit Capital", "Probability of Default (PD)", "Loss-Given-Default (LGD)", "Through-The-Cycle" (TTC), etc. Most banks require all credits to be rated using their own methods and data.

Credit ratings can also be assigned by rating agencies such as Moody's, Standard & Poor (S&P) and Fitch, or by academic researchers in various academic institutions. These researchers study companies' structures and explore mechanics of credit default. They often develop credit models or theories to generate credit ratings that reflect their understanding and theoretical reconstructions.

A quick summary: in today's financial industry, there exist various types of credit ratings created for different purposes (for example compliance or pricing) with different data (for example market public stock data or bank private information) on different customers (for example large companies or small businesses) and from different viewpoints (for example Through-The-Cycle or Point-In-Time) at either individual credit level or portfolio level.

The author has many years of experience in developing credit ratings for multiple major US and world banks where the customers are often large and their compliances often complex. Large companies provide complete and quality financials that are refreshed quarterly. When a credit assessment is assigned to a loan for a large corporation, it often contains two separate components: the likelihood of the company failing and the recovery value of the loan collaterals. The former is closely associated with Probability of Default (PD) and the latter is associated with Loss Given Default (LGD) reflecting the value of collateral.

However, after becoming the Chief Credit Officer of a community bank, the author is more concerned about Small and Medium (SM) businesses as they are our majority customers. It is generally understood by the financial industry that credit ratings for SM loans are more challenging, due to multiple reasons. SM businesses often have less data available. Historical data, if available, may lack consistency or rigor. Rating methods vary among different banks and ratings are often developed by processes that are less quantifiable. Credit ratings produced are hard to be compared with public ratings (such as agency ratings). Furthermore, there is a subtle but extremely important difference between a loan to a large company and a loan to a small business: the former can be unsecured and not dependent on the company's asset structure while the latter could be secured by collaterals that often include all business assets.

2 Credit Rating 2.0 is Needed

When the Basel framework for credit risk management was established, banks quickly realized that they did not have enough historical credit default data to develop credit ratings that meet Basel requirements. The data they had often lacked consistency, transparency and comparability. Only after about a decade the banks were finally able to overcome these challenges and develop credit ratings acceptable for credit risk management and compliance. I call these ratings **Credit Rating 1.0.**

In 2012 after the 2008 financial crisis, the US Security and Exchange Commission (SEC) presented a report (US SEC, 2012) to the US Congress on the "Credit Rating Standardization" requirement by the Dodd-Frank Wall Street Reform and Consumer Protection Act (DFA). The objectives of the report are, among other things, the feasibility and desirability of "standardizing credit rating terminology" and "requiring a quantitative correspondence between credit ratings and a range of default probabilities and loss expectations."

Why did the DFA and the Congress want to standardize credit ratings? What went wrong with Credit Rating 1.0?

During the financial crisis, many issues were identified with credit ratings. First and foremost is the inconsistency issue. There are actually two types of inconsistencies: 1) customers with the same rating assigned by the same bank or institution could behave notably differently; 2) credit ratings by different institutions are not easy to convert, they are not even consistent. Credit conversions are often less stable. I call the first type of inconsistency Cross-Time inconsistency, and the second type of inconsistency Cross-System inconsistency.

The Cross-Time inconsistency can be demonstrated by an example of Moody's ratings. During the financial crisis, the market has observed wide fluctuations in Default Probabilities over one year period, as illustrated by the following Table 1 for Moody's historical B-Rated Corporate Rating and its corresponding historical 1-year default rates. Bank ratings had similar experiences with their internal ratings.

However, as we will see later in this paper, this seemingly problematic behavior unveils an important insight that can help our understanding and advancement of credit ratings. We will show that this is not a problem, but a well-explainable and manageable behavior. In fact, this is what should be expected.

Table 1 Moody's historical B-Rated Corporate Rating Default Rates vary widely.

Moodys B	2001	2002	2003	2004	2005	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016
Historical PD	9.8%	5.4%	2.8%	0.7%	1.3%	0.0%	2.1%	7.6%	0.5%	0.1%	0.5%	0.9%	0.3%	2.3%	1.6%

The Cross-System inconsistency can be best described by mapping the well-known KMV EDF credit ratings into banks internal credit ratings. Most banks have to implement a quarterly "Rating Calibration" process to re-adjust rating conversions periodically. Another example of Cross-System inconsistency is the conversion of credit ratings between two banks, in particularly when a bank acquires another bank. In bank acquisitions, sometimes banks choose to keep the acquired banks' ratings unconverted due to difficulties in converting them.

Other credit rating critics included: ratings were discrete or less granular; rating changes were slow in responding to events; rating methodologies were sometimes less transparent; ratings were sometimes derived with complicated formulae or complex processes, etc.

Why do we have these issues? These issues exist due to multiple reasons including the following.

1. Prior to the financial crisis, many banks stretched to develop credit ratings using limited historical data and simple statistical models. They have not had time to observe and understand the behaviors of these ratings.

2. Important credit information may not be contained in the historical data - for example, credit deteriorations including upgrades and downgrades were not available when the credit ratings were established for the first time. In other words, Credit Rating 1.0. missed some key credit information and behaviors.

3. The essences of creditworthiness and credit rating were not fully understood. In fact, a common misinterpretation by almost all banks and researchers is that credit ratings are the same as Probabilities of Default (PDs) over a period of time (typically one-year). This common misconception is indeed one of the root causes of many issues. A profound understanding of the relationship between credit rating and various PDs (for example TTC PD and PIT PD) is needed.

4. Banks are heavily regulated and regulations have ultimate impacts on the creations and usages of credit ratings. In other words, credit ratings have to reflect key regulatory requirements (such as credit allowance and credit capital). However, Credit Rating 1.0 could not fully incorporate such relationship. For example, a better credit should have a better chance to survive. This implies that credit rating is related to survival probability in extreme situations (which is directly measured by Basel's credit capital).

How do we fix these issues? We need **Credit Rating 2.0**. which should meet at least the following requirements.

- It leverages additional credit information to capture important behaviors of credit rating migrations.
- It is an enhancement to, not a replacement of, the existing ratings. In other words, Credit Rating 2.0 should be an extension of the existing ratings.
- It is consistent across time with good discriminatory power. The rating has good granularity or is continuous.
- It provides a good framework to work with various types of PDs including PD over a time period or PD over the lifetime. In particular, it can unify two seemly different approaches: Through-The-Cycle (TTC) and Point-In-Time (PIT).
- It can be applied to both the Unsecured and Secured ratings.
- It can be used to derive the rating for a customer or the rating for a portfolio. The difference between individual customers and portfolios is the diversification benefit. For example, even every loan in a portfolio is rated A, the portfolio may be rated a bit better (or worse).
- It provides direct connections with key regulatory requirements.
- It provides a measure for the differences between different rating systems and can stably and consistently convert ratings between them.

Our Time-Consistent (TiC) Credit Rating is an example of Credit Rating 2.0. In addition, it is universal in the sense that

- All existing credit ratings (Banks' ratings, Agency Ratings, Academic Ratings, Small Business Ratings, portfolio or individual credit) can be produced using one single mathematical formula.
- Current ratings can be easily extended and enhanced.

Although the original TiC method was developed for Obligor (Unsecured) ratings, we can extend it to SM loans (which will be for Secured instead of Unsecured loans). This requires combining two risk assessments (PD and LGD) into one rating and enables us to assign agency-equivalent ratings to SM loans. We can validate such ratings using historical SBA data,

3 Structures of this paper

Section 4 is a review of Time-Consistent (TiC) Credit Ratings for Agencies, Banks and First-passage model. The relationships between regulatory requirements (Allowance & Capital) and TiC driving factors CCM (or λ) and μ are explained in this section. The same TiC framework is used to deal with both TTC and PIT PDs.

In section 4.6, we exhibit direct and quantifiable relationship between key regulatory requirements (allowance and capital) and credit ratings.

Section 6 introduces non-regulatory arbitrage rating conversions between various credit rating systems, especially conversions from PIT to TTC. We also define credit outlook using PIT and TTC PDs.

4 Review of Time-Consistent (TiC) Credit Rating Approach

In this section, we explain the concepts and main results of TiC Credit Rating.

We can roughly classify the existing credit ratings in the financial industry into three approaches:

Process-Specific Approach is often used by rating agencies (Moody's and S&P, for example) where both quantitative and qualitative information are analyzed through complex "Credit Assessment processes" and rating decisions are made based on the borrower's ranking order of "credit risk". The "Process-Specific" approach does not produce formulae for default probabilities, but historical defaults by ratings (such as Table 1) can be observed. In this approach, credit ratings may not necessarily be equivalent to PDs.

Process-Specific Approach may suffer from Cross-Time inconsistency as it is often difficult to maintain a consistent rating process with appropriate information all time (banks for example, often experience various mergers and acquisitions). Unless it has long historical data, PD calculation is generally difficult. In addition, the rating is often discrete which is less desirable for dynamic credit hedging.

Distribution-Specific Approach is often used by large banks where historical default data are analyzed through mostly Probit or Logistic regressions to produce borrower's ranking order of "default risk" and expected PD values. We call this approach "Distribution-Specific" because in such regressions, the distributional forms of default probabilities are implicitly assumed (for example normal distributions for Probit and Logistic distributions for Logistic regressions). In this approach, credit rating is usually equivalent to PD.

Distribution-Specific approach makes direct assumptions regarding default distributions to generate ranking orders based on the PD values so credit ratings are monotonic functions of default probabilities. However, tying up credit ratings to PD values creates a number of issues: there are many types of PD to choose: 1-year PD, cumulative PD, Point-In-Time (PIT) PD, Through-The-Cycle (TTC) PD with possible different ranking orders. For any selected PD type, its values may vary as well as we have seen in Table 1 (the Cross-Time inconsistency issue).

Default-Specific Approach is often used by academic researchers where default is an event unambiguously defined and mathematically modeled. For example, the first-passage default model that specifies default as the firm's asset value hits a barrier for the first time or KMV Merton structural model where the default can only happen at maturity. One of the main objectives of such approach is to derive theoretical formulae for default probabilities. Credit ratings are often produced using default probabilities.

Default-Specific approach is commonly used by academic researchers for modeling a specific default behavior (for example, the first-passage default in which the default happens when the asset falls below the debt for the first time; or KMV Merton structural model where the default can only happen at the maturity) while default in real world is often determined by many criteria. This approach often incorporates data different from bank data so rating conversion becomes an issue (for example, converting PIT PD to TTC PD). Another issue is rating instability due to the use of market data.

Most banks discuss credit ratings at the individual asset level. However, when they are required to calculate risk capitals (such as Basel's Economic Capital), they need to deal with a very tough issue: the so-called diversification benefit which only exists for portfolios. As a result, a portfolio can perform better collectively than each individual loan in it (so it deserves a better rating). To rating such a portfolio, a good method that incorporates (directly or implicitly) the diversification benefit is needed.

Our Time-Consistent (TiC) Credit Rating takes a different approach: the Behavior-Specific approach in which we differentiate credits by their behaviors: similar credits are expected to have similar behaviors. This approach requires identifying and quantifying credit behaviors. It has a big advantage in the sense that any other credit rating approach must either fit well with essential credit behaviors (hence consistent with TiC approach) or fail to capture important behaviors (hence a poor approach). This makes it possible for TiC to be a "universal" approach. On the other hand, it is quite challenging to define fundamental credit behaviors and to come with appropriate data for their calculations. TiC identifies two key credit behaviors: credit deterioration and credit default. They are described using two mathematical quantities CCM (Credit Corrosion Measure) and μ (Credit Life-expectancy). A TiC Credit Rating is created through a single formula using both quantities and their relationship. In this paper, we will

- Demonstrate both quantities under the three common approaches: Agency ratings, Banks' ratings and First-passage model (or KMV) ratings;
- Extend calculations to Small and Medium business loans that are secured by collaterals;
- Provide TiC rating formulae for aforementioned ratings so they become TiC-enhanced;
- Establish theory and mathematical calculations for various PDs including PIT PD and TTC PD;
- Establish theoretical connections between CCM, μ and two key regulatory requirements: Allowance and Capital;
- Apply the method to both individual loans and portfolios;
- Obtain differences between different rating systems and provide methods for stable rating conversions

The idea behind the TiC approach is the dynamical nature of credit default time. Let τ be the default time of a credit. It changes when credit conditions change. The distribution of τ often has the following shape: in the early ages of a credit, the chance to default is very low. As the age grows, the default chance climbs to its peak. Then it starts to decline (until the credit exits or defaults), as demonstrated by Figure 1. below. This observation enables us to develop the basic concepts and essential components of TiC Credit Rating.

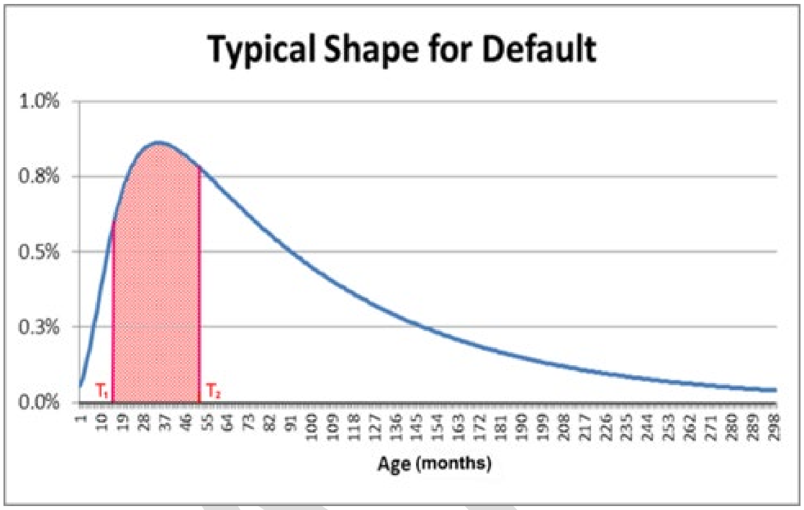

Figure 1. For a credit, its default behavior varies depending on the age.

Credit rating is also dynamical. When the credit condition changes, it most likely changes but sometimes stays unchanged due to potential off-setting effects among various driving factors.

We have identified two credit driving factors: Credit Corrosion factor (CCM) and Life Expectancy factor μ, both are assumed to change constantly.

4.1 Definitions of TiC Credit Rating and the Two Credit Risk Driving Factors

DEFINITION 4.1. (TiC Credit Rating) *For any given credit, let τ > 0 be its default time. The two driving factors are defined as follows.*

1) *The first factor is the Life Expectancy of τ and is defined as*

$$\mu = E[\tau] \tag{1}$$

2) *The second factor is called the Credit Corrosion Measure (CCM) and is defined as*

$$CCM = E[\tau]\cdot E[1/\tau]-1 \tag{2}$$

3) *The Default Peak λ is defined as the mode-to-mean ratio for τ :*

$$\lambda = \frac{the\ Mode\ of\ \ \tau}{E[\tau]} \tag{3}$$

4) *The Time-Consistent (TiC) Credit Rating, for a given constant Q, is defined as*

$$TiC = \frac{CCM}{\mu^{Q}} \tag{4}$$

Q is called Constant of Rating System.

We often use RiskScore (RS) which is 100xTiC instead of TiC.

$$RS = 100 \cdot TiC = 100 \cdot \frac{CCM}{\mu^{Q}} \tag{5}$$

A few things to note: a) CCM measures aggregated credit deterioration beyond expected level; b) both CCM and μ are dynamic. They change as time changes. We require that the percentage change in CCM is proportional to the percentage change in μ in order to produce the same TiC rating, Q is the ratio of the exchange between the two factors; c) Q also measures the differences between different rating systems; d) λ reflects the timing (early or late) of default.

To visualize TiC rating and the two factors, we use a coordinate system (μ-CCM Chart) with ln(μ) being the *x*-axis and ln(CCM) the *y*-axis, as demonstrated by Figure 2.

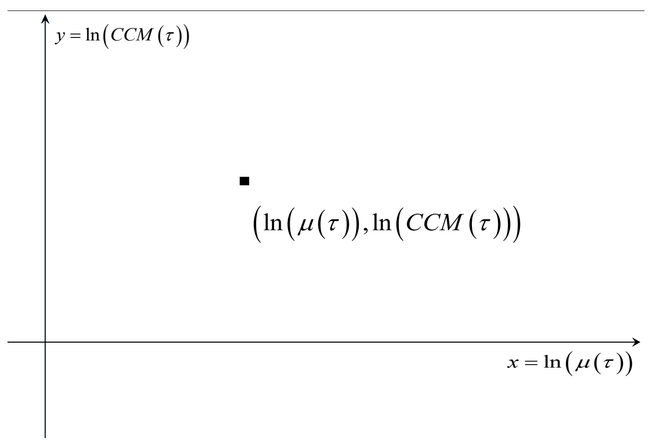

Figure 2. On μ-CCM Chart, credit ratings have coordinates $(\ln(\mu), \ln(CCM))$.

In Figure 3 below, we demonstrate μ-CCM Chart using Moody's historical Caa-C ratings. Each point represents a different year. It shows that the Caa-C ratings are fairly consistent across time as it forms a line with Q being the slope.

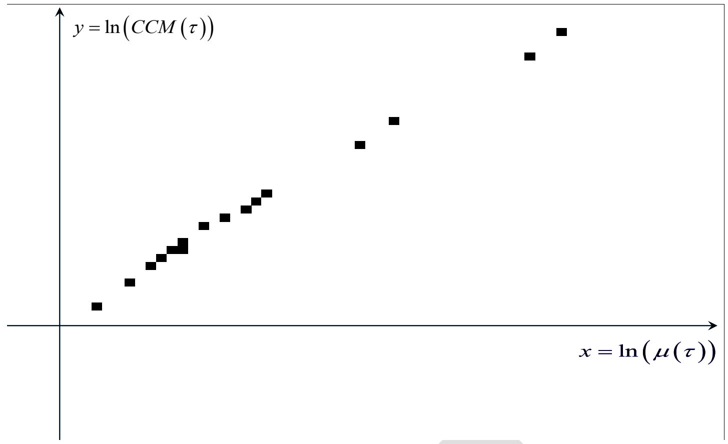

Figure 3. Moody's historical Caa-C ratings are plotted on µ-CCM Chart. Each dot represents a year.

4.2 TiC Ratings for Agency Ratings

We demonstrate the TiC approach for agencies' ratings in this section. The data we use are Moody's published "Annual Default Study: Corporate Default and Recovery Rates" (See Moody's Investors Service (2011), (1920-2010)), and S&P "Annual Global Corporate Default Study And Rating Transitions" (see S&P Global Ratings (2016), (2017)). As both Moody's and S&P have 20+ rating grades, one year is too short for many grades to observe meaningful defaults. We utilize their published rating transition matrices in which several adjacent ratings are combined. Moody's matrices are between 2001 and 2016 (missing 2006) and S&P matrices are between 2005 and 2016 (missing 2008).

These matrices contain a "WR" category for ratings withdrawn before year end. We remove the WR category by re-distributing the probabilities. Also, we calculate the "Average Transition Matrices" that are the averages of the historical transition matrices (See Tables below).

Table 2. Average historical transition matrices for Moody's and S&P.

Moody's Avg	Aaa	Aa	A	Baa	Ba	B	Caa-C	D
Aaa	90.4%	8.9%	0.6%	0.0%	0.0%	0.0%	0.0%	**0.0%**
Aa	1.0%	90.1%	8.4%	0.4%	0.0%	0.0%	0.0%	**0.0%**
A	0.1%	2.8%	90.9%	5.5%	0.5%	0.1%	0.0%	**0.1%**
Baa	0.0%	0.2%	4.8%	89.4%	4.4%	0.8%	0.2%	**0.2%**
Ba	0.0%	0.1%	0.4%	6.2%	83.4%	8.0%	0.7%	**1.2%**
B	0.0%	0.0%	0.1%	0.4%	5.3%	82.2%	7.2%	**4.7%**
Caa-C	0.0%	0.0%	0.0%	0.1%	0.5%	8.7%	71.1%	**19.5%**
D	0%	0%	0%	0%	0%	0%	0%	100%

S&P Avg	AAA	AA	A	BBB	BB	B	CCC/C	D
AAA	90.1%	9.0%	0.6%	0.0%	0.1%	0.0%	0.0%	**0.0%**
AA	0.6%	90.7%	7.9%	0.6%	0.1%	0.1%	0.0%	**0.0%**
A	0.0%	1.9%	91.8%	5.6%	0.4%	0.2%	0.0%	**0.1%**
BBB	0.0%	0.1%	3.8%	91.0%	4.1%	0.7%	0.1%	**0.2%**
BB	0.0%	0.1%	0.2%	5.5%	84.2%	8.5%	0.7%	**0.9%**
B	0.0%	0.0%	0.1%	0.2%	5.3%	84.7%	5.2%	**4.4%**
CCC/C	0.0%	0.0%	0.2%	0.3%	0.9%	13.7%	51.3%	**33.6%**
D	0%	0%	0%	0%	0%	0%	0%	100%

PROPOSITION 4.2. (TiC Credit Ratings for Agency Ratings)

1) *Moody's has Constant of Rating System Q = 0.746, and S&P has Q= 0.626*
2) *Moody's and S&P's CCM, μ and RiskScores are presented in Tables below.*

Table 3. CCM and μ for Moody's and S&P.

Moody's		Aaa	Aa	A	Baa	Ba	B	Caa-C		Aaa	Aa	A	Baa	Ba	B	Caa-C
μ	**2001**	209.3	110.6	89.7	71.6	42.5	14.9	3.3	**2010**	274.0	260.3	251.1	252.6	208.1	134.2	40.3
CCM		0.57	0.95	1.51	2.20	2.91	3.62	2.49		1.23	1.53	1.80	2.05	5.19	8.62	17.65
μ	**2002**	51.7	39.5	30.4	21.0	15.0	9.9	3.6	**2011**	100.7	85.1	81.7	76.5	58.8	40.1	11.6
CCM		0.24	0.34	0.50	0.70	0.75	1.50	2.73		0.45	0.70	0.78	0.98	2.63	2.76	7.14
μ	**2003**	149.1	87.3	60.0	42.0	27.5	17.4	6.4	**2012**	89.1	85.4	82.9	77.5	51.4	25.1	4.7
CCM		1.11	0.57	0.54	0.73	1.19	1.91	4.49		0.66	0.73	0.79	1.00	2.32	2.71	3.47
μ	**2004**	1508.7	1464.2	1432.0	1377.0	1257.1	929.9	448.4	**2013**	147.8	122.8	105.2	89.4	54.4	26.2	4.4
CCM		2.11	4.17	8.69	15.52	24.97	33.95	91.99		0.37	0.54	0.80	1.37	2.20	2.83	3.27
μ	**2005**	588.7	578.6	544.5	503.0	447.0	365.5	287.4	**2014**	333.4	288.3	274.6	237.5	143.0	61.0	10.0
CCM		1.72	1.88	3.64	7.44	12.03	26.75	68.14		0.78	1.34	1.71	3.43	4.37	4.68	6.38
μ	**2007**	758.0	741.0	715.8	621.4	427.4	225.1	61.1	**2015**	168.7	137.9	119.9	96.6	56.6	29.2	8.3
CCM		2.70	3.47	6.27	8.72	10.83	12.58	23.63		0.42	0.66	0.97	1.57	2.08	2.48	5.53
μ	**2008**	67.8	59.9	57.3	43.6	22.7	10.2	2.5	**2016**	116.9	82.3	65.0	46.2	32.9	17.9	4.8
CCM		0.36	0.44	0.59	0.99	1.30	1.03	1.92		0.23	0.32	0.47	0.78	1.29	1.97	3.49
μ	**2009**	37.3	34.4	30.8	25.4	13.3	6.4	1.8	**Avg**	118.7	109.2	98.3	81.4	54.5	30.1	13.7
CCM		0.29	0.33	0.41	0.60	0.80	0.87	1.36	**Matrix**	0.57	0.72	1.01	1.79	2.75	4.25	8.08

S&P		AAA	AA	A	BBB	BB	B	CCC/C		AAA	AA	A	BBB	BB	B	CCC/C
μ	**2005**	158.7	149.7	123.9	102.4	75.0	58.2	42.3	**2012**	362.8	323.1	303.2	266.3	156.7	81.3	28.1
CCM		0.44	0.51	0.84	1.40	2.71	7.15	18.29		0.67	0.96	1.24	2.35	4.13	7.14	13.69
μ	**2006**	581.9	522.0	448.1	383.6	272.8	216.1	100.1	**2013**	1578.5	1487.9	1441.1	1399.2	1213.7	792.1	243.7
CCM		0.95	1.59	3.23	4.29	6.81	12.21	33.23		1.20	2.64	6.70	11.04	19.07	36.14	60.94
μ	**2007**	375.7	316.8	272.4	227.1	183.1	160.4	88.5	**2014**	631.5	627.5	620.3	558.1	430.0	296.3	59.0
CCM		0.61	1.07	1.70	2.66	4.69	9.83	30.53		1.41	1.44	1.50	3.35	6.86	15.31	23.05
μ	**2009**	47.7	35.7	32.5	26.3	14.8	6.6	1.9	**2015**	321.5	247.1	209.9	188.2	89.7	35.2	5.0
CCM		0.24	0.32	0.40	0.64	0.93	1.69	1.44		0.43	0.75	1.37	2.23	3.18	3.66	3.63
μ	**2010**	656.9	653.7	646.1	612.8	503.4	397.8	228.0	**2016**	191.5	188.5	170.3	136.4	81.1	32.1	7.5
CCM		2.01	2.07	2.19	4.50	10.75	28.36	58.25		0.78	0.81	1.19	2.43	3.50	4.48	5.10
μ	**2011**	224.3	223.2	216.4	198.1	153.8	111.5	1.8	**Avg**	108.8	100.6	91.4	76.9	51.7	30.8	12.2
CCM		0.93	0.94	1.04	1.56	3.62	11.58	1.36	**Matrix**	0.51	0.63	0.88	1.50	2.62	4.33	7.41

3) *RiskScores are consistent across time and across ratings, with good discriminatory powers. 1-Year PD does poor separations for good credits.*

Table 4. RiskScores for Moody's and S&P are quite stable/consistent and separate credit very well while PDs don't.

Moody's	Aaa	RS	Aa	RS	A	RS	Baa	RS	Ba	RS	B	RS	Caa-C	RS
2001	0.00%	1.1	0.00%	2.8	0.17%	5.3	0.30%	9.1	1.2%	18	9.8%	48	34.8%	103
2002	0.00%	1.3	0.00%	2.2	0.17%	3.9	1.25%	7.2	1.6%	10	5.4%	27	31.4%	105
2003	0.00%	2.7	0.00%	2.0	0.00%	2.5	0.00%	4.5	1.0%	10	2.8%	23	24.4%	112
2004	0.00%	0.9	0.00%	1.8	0.00%	3.8	0.00%	7.1	0.2%	12	0.7%	21	13.3%	97
2005	0.00%	1.5	0.00%	1.6	0.00%	3.3	0.18%	7.2	0.0%	13	1.3%	33	8.9%	100
2007	0.00%	1.9	0.00%	2.5	0.00%	4.6	0.00%	7.2	0.0%	12	0.0%	22	23.3%	110
2008	0.00%	1.6	0.53%	2.1	0.34%	2.9	0.47%	5.9	1.1%	13	2.1%	18	37.9%	97
2009	0.00%	1.9	0.00%	2.3	0.19%	3.1	0.79%	5.4	2.5%	12	7.6%	22	51.4%	87
2010	0.00%	1.9	0.00%	2.4	0.22%	2.9	0.00%	3.3	0.0%	10	0.5%	22	26.5%	112
2011	0.00%	1.4	0.00%	2.5	0.00%	2.9	0.09%	3.8	0.2%	13	0.1%	18	25.5%	115
2012	0.00%	2.3	0.00%	2.6	0.00%	2.9	0.08%	3.9	0.2%	12	0.5%	25	40.4%	109
2013	0.00%	0.9	0.00%	1.5	0.00%	2.5	0.07%	4.8	0.5%	11	0.9%	25	42.1%	108
2014	0.00%	1.0	0.00%	2.0	0.12%	2.6	0.07%	5.8	0.2%	11	0.3%	22	28.1%	114
2015	0.00%	0.9	0.00%	1.7	0.00%	2.7	0.00%	5.2	0.3%	10	2.3%	20	16.1%	114
2016	0.00%	0.7	0.00%	1.2	0.00%	2.1	0.00%	4.5	0.1%	10	1.6%	23	28.6%	109
Avg	0.00%	1.6	0.02%	2.2	0.06%	3.3	0.19%	6.7	1.2%	14	4.7%	34	19.5%	115

S&P	AAA	RS	AA	RS	A	RS	BBB	RS	BB	RS	B	RS	CCC/C	RS
2005	0.00%	1.9	0.00%	2.2	0.00%	4.1	0.13%	7.7	0.4%	18	2.5%	56	13.1%	175
2006	0.00%	1.8	0.00%	3.2	0.00%	7.1	0.00%	10.4	0.4%	20	0.7%	42	17.5%	186
2007	0.00%	1.5	0.00%	2.9	0.00%	5.1	0.00%	8.9	0.4%	18	0.1%	41	19.4%	184
2009	0.00%	2.1	0.00%	3.4	0.19%	4.6	0.58%	8.3	0.9%	17	11.0%	52	57.3%	96
2010	0.00%	3.5	0.00%	3.6	0.00%	3.8	0.00%	8.1	0.0%	22	1.3%	67	25.2%	195
2011	0.00%	3.1	0.00%	3.2	0.00%	3.6	0.14%	5.7	0.0%	15	1.8%	61	18.3%	94
2012	0.00%	1.7	0.00%	2.6	0.00%	3.5	0.00%	7.1	0.0%	17	1.2%	46	33.7%	169
2013	0.00%	1.2	0.00%	2.7	0.00%	7.1	0.00%	11.8	0.0%	22	1.0%	55	33.3%	195
2014	0.00%	2.5	0.00%	2.6	0.00%	2.7	0.00%	6.4	0.0%	15	0.6%	43	33.3%	180
2015	0.00%	1.2	0.00%	2.4	0.00%	4.8	0.00%	8.4	0.2%	19	2.7%	39	36.7%	133
2016	0.00%	2.9	0.00%	3.0	0.00%	4.8	0.00%	11.2	0.6%	22	4.2%	51	46.0%	144
Avg	0.00%	2.7	0.03%	3.5	0.07%	5.2	0.23%	9.9	0.9%	22	4.4%	51	33.6%	155

4) *TiC Credit Rating is consistent in each year, even when PD is 0. For example, RiskScores for B-rated are consistent with little fluctuation.*

Table 5 RiskScore for Moody's B-Rated are fairly consistent across time.

Moodys B	2001	2002	2003	2004	2005	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016
Historical PD	9.8%	5.4%	2.8%	0.7%	1.3%	0.0%	2.1%	7.6%	0.5%	0.1%	0.5%	0.9%	0.3%	2.3%	1.6%
RiskScore	48	27	23	21	33	22	18	22	22	18	25	25	22	20	23

5) *In Figure 4 below, RiskScores represent the y-intercepts of grade lines. The slope is Q.*

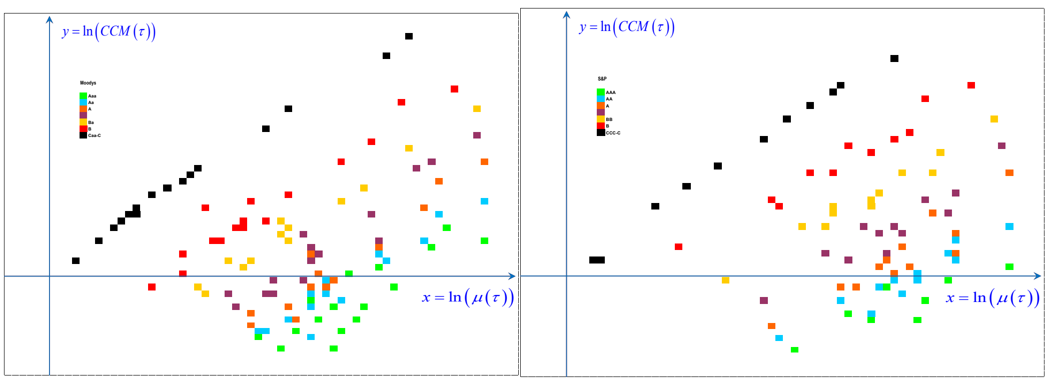

Figure 4. Each data point represents a year and each color represents a rating. The left figure is for Moody's and the right figure is for S&P. Q is the slope of the lines.

To understand the relationship between PD and credit rating, we add default probabilities (the 1-year PD) to the μ-CCM chart in Figure 4. To do so, one needs to know the distribution of τ for each rating grade. For demonstration purposes, in Figure 4 and Figure 5 below, we assume Log-normal distributions for all grades and use Equation (6) to calculate PDs. As it shows (some numbers are too small to be recognized) that directions of the 1-year PDs are largely consistent with (but slightly different than) credit grades (or rating lines). This is the reason why we observe fluctuations in historical PDs. This can be better visualized using the "Heat Map" in Figure 5 where different PDs are represented by different colors.

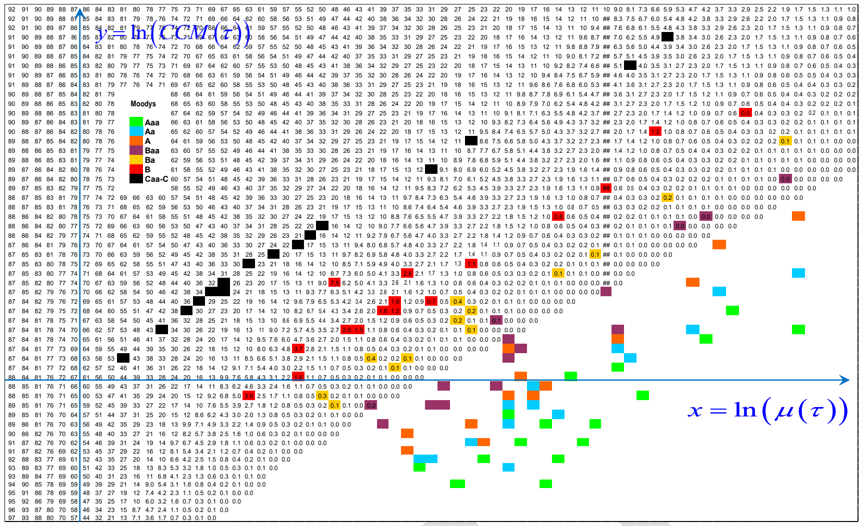

Figure 5. PDs for Moody's ratings (any PD less than 0.1bp is left blank).).

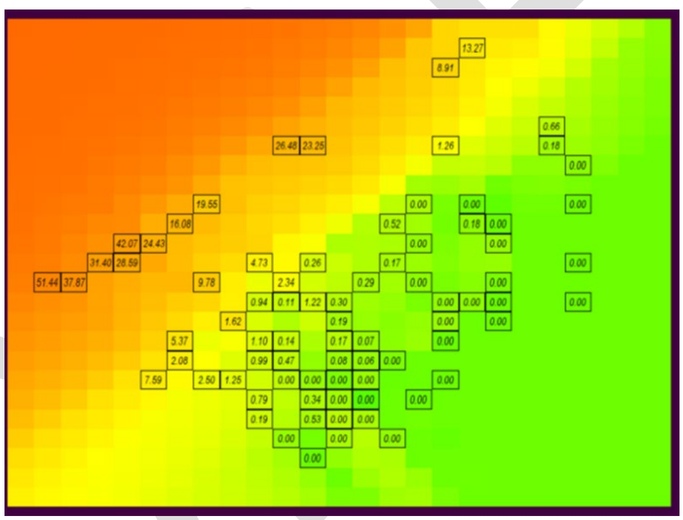

Figure 6. Moody's PD "Heat Map": same PDs are represented by the same color. Each box is a grade in a year and the numbers in the boxes are 1-year default probabilities (%).

In Figure 5 we see that 1-year PDs vary along the rating lines, although their values generally increase as the rating lines moving up (the rating getting worse). This is because 1-year PD is not exactly Time-consistent.

4.3 TiC Credit Ratings for Bank Ratings

Banks keep their customer rating data strictly confidential. However, in my many years credit experience with many large banks (US and world), I have observed a common practice in most banks in establishing their internal credit ratings: nearly all banks use either Probit or Logistic regressions to estimate PDs before they are "calibrated" to credit ratings. In these regressions, the credit age T often serves as a vital exogenous variable. In fact, the new CECL/IFRS 9 regulations require banks to estimate losses based on credit ages. Our observation is: choosing different regressions is the same as selecting different distributions for default time τ. Hence, we have the following results regarding banks credit rating models.

PROPOSITION 4.3.1. (Banks' TiC Credit Rating Models) *If the default probability is modeled using Probit (or Logistic) regression and the logarithm of T is included in the model as an exogenous variable, then the default time τ follows a Log-normal (or Log-logistic) distribution. Their Time-Consistent ratings (equivalently, E[τ] and E[1/τ]) can be calculated directly using these two distributions. Let Φ be the Cumulative Distribution Function (cdf) of the standard normal distribution and λ the Default Peak. Then we have the following.*

1) *For the Probit regression model,*

$$\begin{cases} CCM = \lambda^{-\frac{2}{3}} - 1 \\ PD_T = \Phi\left(\dfrac{\ln T - \ln \mu + \frac{1}{2}\ln\left(CCM+1\right)}{\sqrt{\ln\left(CCM+1\right)}} \right) \end{cases} \tag{6}$$

Here PD_T is the default probability in the first T years.

2) *For the Logistic regression model,*

$$\begin{cases} CCM = \dfrac{\left(\pi s\right)^2}{\sin^2\left(\pi s\right)} - 1 \qquad\qquad 0 < s < 1 \\ \lambda = \left(\dfrac{1-s}{1+s}\right)^s \dfrac{\sin\left(\pi s\right)}{\pi s} \\ PD_T = \dfrac{1}{1 + e^{\frac{-\ln T + \ln \mu - \frac{1}{2}\ln\left(CCM+1\right)}{s}}} \end{cases} \tag{7}$$

3) *Both PD_T and CCM are decreasing functions of λ.*
4) $\lim_{\lambda \to 0} CCM = \infty$ *and* $\lim_{\lambda \to 1} CCM = 0$.

For a known distribution (such as Log-normal or Log-logistic), we can establish an explicit connection (to be called Regulatory Relationship) between its credit ratings and its two observable behavioral parameters: T-year PD_T and default peak λ. Proposition 4.3.2 below presents this result for both Log-normal distribution and Log-logistic distributions. Such formulae are important because they enable us to calibrate credit ratings directly to real world data without actually running regressions using historical data, PROPOSITION 4.3.2. (TiC Credit Rating and Regulatory Relationship I)

1) *For Probit Regression Model, we have the following formulae for rating, PD and default peak λ (or CCM),*

$$\ln(TiC) = Q\cdot\Phi^{-1}(PD_T)\cdot\sqrt{-\frac{2}{3}\ln(\lambda)} + \frac{Q}{3}\cdot\ln(\lambda) + \ln\left(\frac{1}{\lambda^{\frac{2}{3}}} - 1\right) - Q\cdot\ln(T)$$
$$= Q\cdot\Phi^{-1}(PD_T)\cdot\sqrt{\ln(CCM+1)} - \frac{Q}{2}\ln(CCM+1) + \ln(CCM) - Q\cdot\ln(T) \tag{8}$$

2) *For Logistic Regression Model, we have*

$$\ln(TiC) = Q\cdot s\cdot\ln\left(\frac{PD_T}{1-PD_T}\right) + Q\cdot\ln\left(\frac{\sin(\pi s)}{\pi s}\right) + \ln\left(\frac{(\pi s)^2}{\sin^2(\pi s)} - 1\right) - Q\cdot\ln T \tag{9}$$

In late section, CCM will be connected to another important regulatory requirement: the credit capital and express a credit rating as an explicit function of PD (Credit Loss Allowance) and Credit Capital requirement. It can be used to measure regulation impacts on credit ratings directly.

PROPOSITION 4.3.3. (TiC Credit Rating for Portfolio) *For a credit portfolio, we can estimate its Life Expectancy μ, as well as it Default Peak λ (hence CCM) using the relationship between λ and CCM in Equations (6) and (7).*

Where is the diversification benefit for the portfolio? It is reflected by the default perk λ which represents a collective behavior of all credits. In fact, it represents the speed of credit deterioration. Early perks have worse ratings than later perks. Later in this paper, we will see λ also determines the capital requirement.

4.4 Time-Consistent Ratings for First-passage Model Ratings

The TiC Credit Rating for first passage model has a big advantage: it stays the same under either Risk Neutral or Empirical default probability, as demonstrated by the next Proposition.

PROPOSITION 4.4.1. (First-Passage Asset Default Model) *Let*

$$A_t = A_0 e^{\left(\eta - \frac{\sigma_A^2}{2}\right)t + \sigma_A W_t}. \tag{10}$$

be the asset process of a company and D its debt. The firm defaults when its asset A_t falls below the debt D for the first time. That it, its default time is given by $\tau = \inf\{t \geq 0:\ A_t \leq D\}$.

Assume $\left|\eta - \sigma_A^2/2\right| > 0$ *and* $D < A_0$ *, then the first-passage default process τ has the following properties.*

1) *τ follows Inverse Gaussian Distribution.*
2) *The Two Driving Factors are*

$$\begin{cases} CCM = \dfrac{\sigma_A^2}{\ln\left(A_0/D\right)\cdot\left|\eta - \sigma_A^2/2\right|} \\ \mu = \dfrac{\ln\left(A_0/D\right)}{\left|\eta - \sigma_A^2/2\right|} \end{cases} \tag{11}$$

3) *The only Time-Consistent Credit Rating which is independent of η is when Constant of Rating System Q =1 with*

$$TiC = \frac{CCM}{\mu} = \frac{\sigma_A^2}{\ln^2\left(A_0/D\right)}. \tag{12}$$

Since it is independent of η, it is invariant under Girsanov transformations. In particular, it stays unchanged under either the Risk-Neutral default probability or Empirical default probability.

4) *For the first-passage model, the default probability is given by*

$$PD_T = \Phi\left(\sqrt{\frac{1}{CCM}}\left(\frac{\sqrt{T}}{\sqrt{\mu}} - \frac{\sqrt{\mu}}{\sqrt{T}}\right)\right) + e^{\frac{2}{CCM}}\Phi\left(-\sqrt{\frac{1}{CCM}}\left(\frac{\sqrt{T}}{\sqrt{\mu}} + \frac{\sqrt{\mu}}{\sqrt{T}}\right)\right). \tag{13}$$

5) *The KMV Distance-to-Default (DD) rating and EDF, defined below, are functions of our two credit driving factors μ and CCM (and we will see they are NOT time-consistent in late sections)*

$$\begin{cases} DD_T = \dfrac{\ln\left(\dfrac{A_0}{D}\right) + \left(\eta - \dfrac{\sigma_A^2}{2}\right)T}{\sigma_A\sqrt{T}} \\ \quad = \dfrac{1 \pm T/\mu}{\sqrt{T}\sqrt{CCM/\mu}} \\ EDF = \Phi^{-1}\left(-DD_T\right) \end{cases} . \tag{14}$$

The only Time-Consistent rating for first-passage model is Equation (12). This rating is invariant under Girsanov transformations and becomes stable under either the empirical return/default probability or the risk neutral return/default probability, as demonstrated by the next Proposition.

When an asset becomes tradable on the market, it is exposed to Market Price Risk (MPR). Its expose to credit risk can be changed. Following the notations for the first-passage default model in the above Proposition, these changes can be captured through the changes in credit risk driving factors.

PROPOSITION 4.4.2. (Interactions Between Market Risk and Credit Risk) *Let r be the risk-free interest rate. Suppose that* $\left(\mu, CCM\right)$ *and* $\left(\mu_{RN}, CCM_{RN}\right)$ *are the driving factors under the empirical and the risk-neutral valuation processes, respectively.* $TiC = \sigma_A^2 \Big/ \ln^2\left(A_0/D\right)$ *is the TiC credit rating defined by Equation (11) and* $MPR = \dfrac{\eta - r}{\sigma_A}$ *is the Market Price of Risk. If* $\eta - \dfrac{\sigma_A^2}{2} \geq 0$ *and* $r - \dfrac{\sigma_A^2}{2} \geq 0$*, then we have the following*

$$\begin{cases} \dfrac{1}{\mu} - \dfrac{1}{\mu_{RN}} = \sqrt{TiC} \cdot MPR \\ \dfrac{1}{CCM} - \dfrac{1}{CCM_{RN}} = \dfrac{1}{\sqrt{TiC}} \cdot MPR \end{cases} . \tag{15}$$

In particular, the TiC credit rating stays unchanged

$$TiC = \frac{CCM_{RN}}{\mu_{RN}} = \frac{CCM}{\mu} . \tag{16}$$

4.5 Credit Capital Requirement and CCM

The focus of this subsection is to reveal a surprising connection between CCM and Capital required by Basel II capital framework (See Basel (2006). It is surprising because Capital requirement is often not incorporated in banks' internal credit rating modeling processes.

What is capital? Capital is a cushion for absorbing excessive losses before a bank becomes insolvent. Since there is in general no caps for bank's potential liabilities and losses, this cushion, no matter how high, cannot guarantee 100% safety for the bank. From regulatory perspectives (to protect consumers and to promote public confidence), this cushion has to be high enough so most large losses can be absorbed. In other words, regulators require the bank to guarantee safety and financial solvency at least, say 99.9%, of the time. This 99.9% is called the (required) confidence level. This requirement can be translated into a requirement on bank's lifespan: the bank is required to survive at least 99.9% of the time. Longer lifespan means higher survival probability. If one considers the range of possible lifespans of a bank, then the required lifespan τ^* for the bank has to be longer than 99.9% of the lifespans in the range.

The key connection to be revealed between CCM and Capital requirement is due to a question raised frequently by the author during his 20+ years credit risk management practices for large banks: why (and how) does a credit rating determine the confidence level of bank's capital? In practice, banks always compare historical average PDs with credit ratings. Capital, as a measure for unexpected excessive losses, is generally not incorporated in the credit rating processes. However, when a bank calculates its Economic Capitals (EC) such as the IRB capital under the Advanced Mathematical Approach by Basel II Accord, its credit rating becomes critical: it determines the confidence level of the calculations and dominates the calculation outcomes. For example, AAA-rated banks often use a confidence level between 99.95% and 99.97% and A-rated banks may use a level between 97.5% and 99%. There are no explicit rules or formulae for determining the confidence levels but all agrees that they depend on bank's (desired) credit ratings.

Tying up capital level with credit rating is an essential regulatory requirement. Inevitably, such requirement will have impacts on credit ratings and rating behaviors, although the current rating processes seem ignore such impacts. The idea behind the CCM-Capital connection can be explained as follows. TiC approach concerns the survival time (or default time) of a credit. Good credits survive long and bad credits survive short. Each credit grade has a "minimal survival time" requirement. It is natural to make the "minimal survival time" proportional to the quality of credit.

The "minimal survival time" can be described as a critical value for the credit lifespan to reach. Mathematically, it can be described as a Value-at-Risk of the credit life distribution τ.

DEFINITION 4.5.1. (Required Survival Time and Confidence Level) *Suppose τ is the default time distribution. Let $0 < \alpha < 1$ be a number (called the confidence level).*

1) *The Required Minimal Survival Time (RMST) is proportional to credit quality. That is, there is a constant C (called the Constant of Minimal Lifespan, CML) such that*

$$RMST = CML \cdot \frac{1}{\left(\frac{CCM}{\mu}\right)}. \tag{17}$$

2) *The Value-at-Risk (VaR) of τ corresponding to the confidence level α is a value* τ^* *such that* $\text{Prob}\left\{\tau \le \tau^*\right\} = \alpha$ *. We write* $\tau^* = VaR_\alpha\left(\tau\right)$ *. That is,*

$$\text{Prob}\left\{\tau \le VaR_\alpha\left(\tau\right)\right\} = \alpha\ . \tag{18}$$

3) *The Required Confidence Level is the confidence level α that matches the RMST:*

$VaR_\alpha = RMST$ *, or equivalently,* $VaR_\alpha \cdot \left(\frac{CCM}{\mu}\right) = CML$ (19)

One of our key observations is that the factor CCM is the sole driver for the confidence level (so capital level is independent of μ. This is consistent with banks' practice in which expected losses are excluded from capitals), as demonstrated by the next Proposition. In fact, we shall see that CCM is solely determined by the default peak λ. In real world, default peak is often associated with default clustering. This means that the capital level is solely determined by the default peak/deterioration/clustering.

PROPOSITION 4.5.2. (Formulae for Credit Capital Confidence Level α) *Let* $\theta = 1/Q > 1/2$. *Suppose that the Capital Confidence Level* $\alpha\left(\tau\right)$ *is required as*

$$VaR_{\alpha(\tau)}\left(\tau\right) \cdot \frac{CCM^\theta\left(\tau\right)}{\mu\left(\tau\right)} = \text{Constant } CML \tag{20}$$

1) *Let* F_τ *be the Cumulative Distribution Function of τ. Then* $\alpha\left(\tau\right)$ *can be calculated as*

$$\alpha\left(\tau\right) = F_\tau\left(CML \cdot \frac{\mu\left(\tau\right)}{CCM^\theta\left(\tau\right)}\right) \tag{21}$$

2) *In particular, we have, for bank ratings or first-passage ratings,*

$$\begin{cases} \alpha = \Phi\left(\dfrac{\ln CML - \theta \ln CCM + \ln\left(CCM+1\right)/2}{\sqrt{\ln\left(CCM+1\right)}} \right) & if\ \ \tau\ is\ Log\text{-}Normal \\ \alpha = \dfrac{1}{1+e^{-\left(\ln CML - \theta \ln CCM + \frac{1}{2}\ln\left(CCM+1\right)\right)/s}} & if\ \ \tau\ is\ Log\text{-}Logistic \\ \alpha = \Phi\left(\dfrac{\sqrt{CML}}{\sqrt{CCM^{\theta+1}}} - \dfrac{\sqrt{CCM^{\theta-1}}}{\sqrt{CML}} \right) + & \\ \quad + \ e^{\frac{2}{CCM}} \Phi\left(-\dfrac{\sqrt{CML}}{\sqrt{CCM^{\theta+1}}} - \dfrac{\sqrt{CCM^{\theta-1}}}{\sqrt{CML}} \right) & if\ \ \tau\ is\ Inverse\ Gaussian \end{cases} \tag{22}$$

3) *The confidence level α=α(CCM) is a decreasing function of CCM and is independent of μ. Smaller λ (or early default peak) implies higher CCM, or lower capital level requirement, and worse credit; and vice versa.*
4) *The limits* $\lim_{CCM \to 0} \alpha\left(CCM\right) = 1$ *and* $\lim_{CCM \to \infty} \alpha\left(CCM\right) = 0.$
   In particular, for any $0 < \alpha^* < 1$*, there is a unique* $CCM^* > 0$ *such that*

$$\alpha\left(CCM^*\right) = \alpha\ . \tag{23}$$

The CCM^* in Equation (22) can be used to produce "Regulatory Arbitrage-Free Credit Rating Conversion".

Both CML and θ>1/2 are important quantities that create flexibilities for credit rating management and prudential regulations (we take θ=1 in this paper). Large CML implies high capital requirements for all. Large θ penalizes good credit while small θ penalizes poor credit.

PROPOSITION 4.5.3. (Agency CMLs) *Agency ratings fit well with ϑ=1 and* $CML = e^{1.35} = 3.857$ *, as demonstrated by Table 6 below.*

Table 6 Capital confidence levels using $CML = e^{1.35} = 3.857$.

Moody's	Aaa	Aa	A	Baa	Ba	B	Caa-C
	99.93%	99.60%	97.84%	89.74%	80.75%	71.50%	59.66%
λ	0.51	0.44	0.35	0.22	0.14	0.08	0.04
S&P	AAA	AA	A	BBB	BB	B	CCC/C
	99.97%	99.84%	98.81%	92.89%	81.78%	71.14%	61.11%
λ	0.54	0.48	0.39	0.25	0.14	0.08	0.04

Combining Proposition 4.3.2 and Equation (21), we obtain the following explicit relationship between TiC Credit Ratings and Regulatory Requirements for probit regressions. Similar relationship exists for logistic regressions.

PROPOSITION 4.5.4. (TiC Credit Rating and Regulatory Relationship II) (For *Log-normal distribution) Let α be the Credit Capital confidence Level, PD_1 the first-year default probability, then*

$$\begin{cases} \ln\left(TiCC\right) = Q \cdot \Phi^{-1}\left(PD_1\right) \cdot \sqrt{\ln\left(CCM+1\right)} - \frac{Q}{2}\ln\left(CCM+1\right) + \ln(CCM) \\ \alpha = \Phi\left(\dfrac{1.35 - \ln CCM + \ln\left(CCM+1\right)/2}{\sqrt{\ln\left(CCM+1\right)}} \right) \end{cases} \tag{24}$$

One can use this formula to directly evaluate the impacts of Regulatory Requirements on credit ratings.

We can also derive the relationship between capital weights and credit ratings. Recall that in Definition 4.5.1, the Required Minimal Survival Time (RMST) is assumed to be proportional to the credit rating. In other words, higher-rated credits are expected to survive longer, given the same amount of capital. Equivalently, this implies that the required capital level should be inversely proportional to the credit rating.

PROPOSITION 4.5.5. (Capital Weights by TiC Credit Ratings) *Capital weights are inversely proportional to Time-Consistent credit ratings. That is, if CW_1 and CW_2 are the required capital weights of two ratings TiC_1 and TiC_2, respectively, then*
Let α be the Credit Capital confidence Level, PD_1 the first-year default probability, then

$$\frac{CW_1}{CW_2} = \frac{TiC_2}{TiC_1} \tag{25}$$

For example, a credit of rating 3.90 would be required to have $\frac{3.90}{4} = 97.5\%$ of the capital required for a credit of rating 4.0.

4.6 The Relationship between Regulatory Requirements and Time-Consistent Credit Ratings: A Trade-Off between Short-Term and Long-Term Interests

Time-Consistent credit ratings are based on two dynamic factors: μ and CCM. Proposition 4.5.2 shows that capital requirements (representing long-term solvency considerations) are determined solely by CCM. In contrast, loan loss reserves (reflecting short-term loss considerations) decrease as μ increases.

Long-term and short-term considerations are inherently interconnected. The Time-Consistent credit rating framework explicitly accounts for the trade-off between these short-term and long-term factors, requiring a balance between potential gains and losses over time. The following figure illustrates the relationship between capital requirements and loss reserve requirements within the μ–CCM coordinate graph.

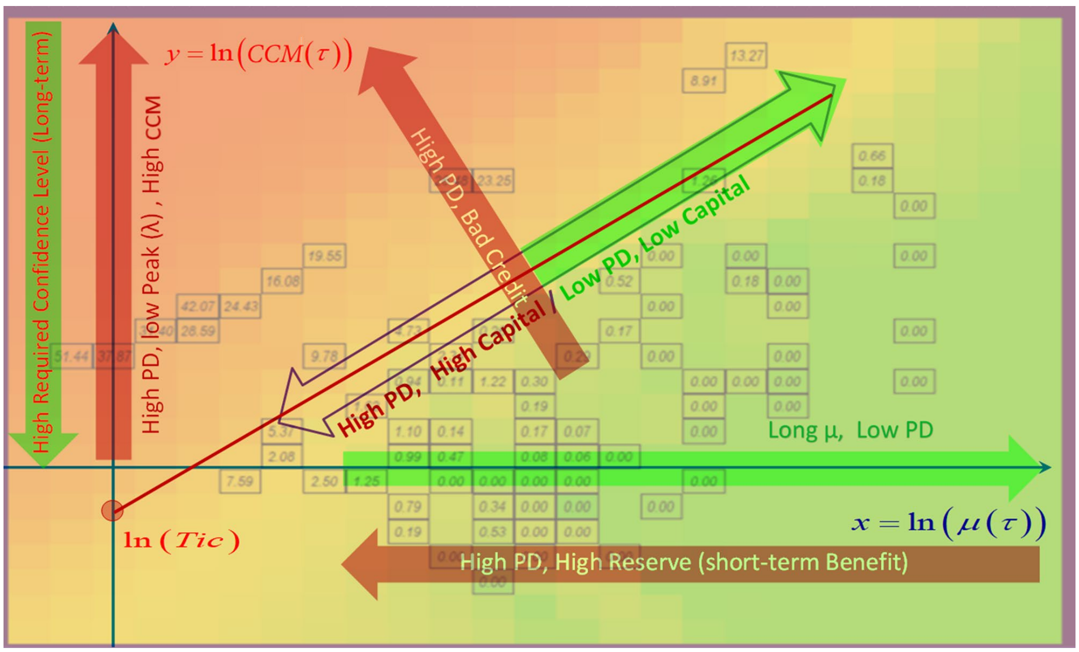

Figure 7. Regulatory requirements on short-term and long-term benefits trade-offs

In Figure 7, colors represent default probabilities (darker colors indicate higher default probabilities). The lines are rating directions: the same line represents the same rating. The higher the position of the line, the worse the rating. On the x-axis, the further to the right, the larger μ, the smaller PD, and the better the rating. Therefore, loss reserve requirements decrease, but the confidence level α increases, and capital requirements increase. These changes can be calculated precisely. On the y-axis, the higher up, the larger CCM, so the confidence level α decreases. This indicates that capital requirements decrease, color becomes darker, PD increases, and loss reserve requirements increase.

In Figure 7, colors represent probabilities of default, with darker colors indicating higher default probabilities. The lines in the figure represent rating directions, where each line corresponds to the same credit rating; lines located higher in the graph correspond to worse ratings.

Along the x-axis, moving to the right indicates a larger value of μ, which corresponds to a lower probability of default and therefore better credit quality. As a result, loss reserve requirements decrease. At the same time, the confidence level α increases, leading to higher capital requirements. These changes can be calculated precisely within the model.

Along the y-axis, moving upward corresponds to a larger value of CCM. In this case, the confidence level α decreases, implying lower capital requirements. At the same time, the color becomes darker, indicating a higher probability of default and therefore higher loss reserve requirements.

Figure 7 quantifies credit ratings, loss reserve requirements, and capital requirements on a μ-CCM coordinate system, depicting Proposition 4.4.

5 Time-Consistent Credit Rating Used to Convert Credit Ratings of Different Rating Systems

Banks often find that converting credit ratings from one rating system to another is not straightforward, and the resulting conversions can sometimes be unstable. We now understand that this instability may arise from differences in the system constant Q across rating systems. The parameter Q represents the slope of the rating line, and for a conversion to be smooth and stable, the slopes of the rating lines in the two systems must be broadly consistent.

In the following sections, we examine the conversions of credit ratings across different rating systems.

5.1 Conversion Between Credit Ratings of Different Agencies

Credit ratings assigned by rating agencies are not generated through explicit formulas but are instead determined through expert judgment and processes. Nevertheless, our analysis shows that the system constants Q underlying these rating systems are quite similar, and the corresponding rating lifespans generally follow a log-normal distribution.

Using the historical average transition matrices of Moody's Investors Service and S&P Global Ratings (Table 2), we can derive their “average” Time-Consistent credit ratings. This can be interpreted as a conversion from the agencies’ Through-The-Cycle (TTC) default rates $\underline{PD}_1$.

PROPOSITION 5.1 (Through-The-Cycle Credit Rating Conversion between Moody’s and S&P) *By comparing their Through-The-Cycle default probabilities and keeping their TTH $\underline{PD}_1$ consistent, we obtain*

1) *Moody's Time-Consistent credit ratings with TTC $\underline{PD}_1$ are shown in the table below*

Moody's	Aaa	Aa	A	Baa	Ba	B	Caa-C
$\underline{PD}_1$	0.01%	0.02%	0.06%	0.19%	1.22%	4.73%	19.55%
μ (Yr)	118.7	109.2	98.3	81.4	54.5	30.1	13.7
CCM	0.57	0.72	1.01	1.79	2.75	4.25	8.08
λ	0.51	0.44	0.35	0.22	0.14	0.08	0.04
RiskScore	**1.6**	**2.2**	**3.3**	**6.7**	**13.9**	**33.5**	**114.7**

Table 7 Moody's Through-The-Cycle $\underline{PD}_1$ and Time-Consistent Credit Rating

2) *S&P's Time-Consistent credit ratings with TTC $\underline{PD}_1$ are shown in the table below*

S&P	AAA	AA	A	BBB	BB	B	CCC/C
$\underline{PD}_1$	0.01%	0.03%	0.07%	0.23%	0.88%	4.41%	33.59%
μ (Yr)	108.8	100.6	91.4	76.9	51.7	30.8	12.2
CCM	0.51	0.63	0.88	1.50	2.62	4.33	7.41
λ	0.54	0.48	0.39	0.25	0.14	0.08	0.04
RiskScore	**2.7**	**3.5**	**5.2**	**9.9**	**22.2**	**50.7**	**154.8**

Table 8 S&P Through-The-Cycle $\underline{PD}_1$ and Time-Consistent Credit Rating

3) *Conversions between Moody's and S&P's credit ratings using TTC $\underline{PD}_1$ (as their system constants Q are similar)*

RiskScore	Aaa	Aa	A	Baa	Ba	B	Caa-C
Moody's	1.6	2.2	3.3	6.7	13.9	33.5	114.7
S&P	2.70	2.95	4.89	9.15	27.4	54.1	127
$\underline{\mathbf{PD}}_1$	0.00%	0.02%	0.06%	0.19%	1.20%	4.70%	19.50%

Table 9 Moody's to S&P Credit Rating Conversion

RiskScore	AAA	AA	A	BBB	BB	B	CCC/C
S&P	2.7	3.5	5.2	9.9	22.2	50.7	154.9
Moody's	1.60	2.61	3.75	7.45	12.8	32.6	140
$\underline{\mathbf{PD}}_1$	0.00%	0.03%	0.07%	0.23%	0.90%	4.40%	33.60%

Table 10 S&P to Moody's Credit Rating Conversion

For example, Moody's Baa (6.7) is converted to 9.15 in the Standard & Poor's, which is slightly better than BBB (9.9).

5.2 No Regulatory-Arbitrage Conversion of Credit Ratings

In general, credit rating conversions should avoid regulatory arbitrage, particularly because the Time-Consistent rating framework is directly linked to two key regulatory requirements: loan loss reserves (PD) and credit capital (CCM).

To convert a credit rating without creating regulatory arbitrage, both the associated loss reserve (PD) and the capital confidence level (α) should remain unchanged. According to Proposition 4.5.2, the confidence level α is a decreasing function of CCM. Therefore, to preserve the same α during the conversion, the rating must be matched with an appropriate CCM in the new rating system. Once the PD and the corresponding CCM are determined, they jointly generate the required credit rating in the new system. This conversion procedure is summarized in the following proposition.

PROPOSITION 5.2.1. (No-Regulatory Arbitrage Credit Rating Conversion)
Suppose that we have two credit rating systems A and B. Assume that

1) *The confidence levels of A and B are given by, as the functions of risk factor CCM, $\alpha = CL_A(CCM)$ and $\alpha = CL_B(CCM)$, respectively.*

2) *The Time-Consistent credit rating TiC_B of system B is given as a function of PD and CCM: $TiC_B = TiC_B(PD, CCM)$.*

Then the following procedure will convert credit ratings of system A into credit ratings of system B without regulatory arbitrage.

1) *Suppose RS_A is a credit rating in system A. It has the probability of default PD_A and the risk factor CCM_A. We solve the following equation for the desired CCM* in system B (in order to keep α the same):*
$$CL_B\left(CCM^*\right) = CL_A\left(CCM_A\right) \tag{26}$$

2) *Substitute PD_A and CCM* into the TiC_B rating equation to obtain*
$$RS_B = TiC_B\left(PD_A, CCM^*\right) \tag{27}$$

Then RS_B is the No-Arbitrage conversion of RS_A.

We now demonstrate this procedure by converting credit ratings between Moody's Investors Service and S&P Global Ratings without having regulatory arbitrage.

Since both rating systems are approximately consistent with log-normal distributions, we adopt the following confidence-level functions derived under the log-normal assumption (see Proposition 4.5.2).

$$\alpha = CL_{MD}\left(CCM\right) = \Phi\left(\frac{1.35 - \frac{1}{0.7462}\cdot \ln CCM + \ln\left(CCM+1\right)/2}{\sqrt{\ln\left(CCM+1\right)}}\right)$$

$$\alpha = CL_{SP}\left(CCM\right) = \Phi\left(\frac{1.35 - \frac{1}{0.625913}\cdot \ln CCM + \ln\left(CCM+1\right)/2}{\sqrt{\ln\left(CCM+1\right)}}\right)$$

Furthermore, Standard & Poor's rating is calculated as, under log-normal condition (see Proposition 4.5.4)

$$\ln\left(TiC_{SP}\right) = 0.625913\cdot\Phi^{-1}\left(PD\right)\cdot\sqrt{\ln\left(CCM+1\right)} - \frac{0.625913}{2}\ln\left(CCM+1\right) + \ln(CCM)$$

It should be noted that the log-normal distribution serves only as an approximation for S&P ratings, since rating agencies do not determine ratings or default rates using explicit formulas. As a result, the theoretical default probabilities derived from formula (6) in Proposition 1.3 differ somewhat from the actual observed one-year default probabilities (PD_1). For example, Moody's ratings exhibit the following discrepancies.

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Moody's	Aaa	Aa	A	Baa	Ba	B	Caa-C
PD_1	0.01%	0.02%	0.06%	0.19%	1.22%	4.73%	19.55%
Log-Normal PD	0.00%	0.00%	0.00%	0.01%	0.19%	2.28%	15.41%
RiskScore	1.6	2.2	3.3	6.7	13.9	33.5	114.7

Table 11 Comparison of Moody's historical default probabilities and the default probabilities under Log-normal assumption

For simplicity, we use the log-normal default probability in the conversion. This approach also avoids potential issues that arise when the observed default probability is zero.

PROPOSITION 5.2.2. (No-Regulatory Arbitrage Conversion for Agency Ratings)

Moody's	Aaa	Aa	A	Baa	Ba	B	Caa-C
CCM	0.57	0.72	1.01	1.79	2.75	4.25	8.08
α	99.98%	99.75%	97.82%	85.81%	71.54%	57.36%	40.75%
RiskScore	**1.61**	**2.16**	**3.29**	**6.70**	**13.93**	**33.53**	**114.72**
S&P	AAA	AA	A	BBB	BB	B	CCC/C
CCM*	0.60	0.75	1.01	1.63	2.29	3.17	4.94
α	99.98%	99.75%	97.82%	85.81%	71.54%	57.36%	40.75%
RiskScore	**2.81**	**3.78**	**5.72**	**11.33**	**21.73**	**45.43**	**120.81**

Table 12 No-Regulatory Arbitrage Conversion from Moody's to S&P

By comparing the Through-The-Cycle rating conversion (Table 9) with the No-Regulatory-Arbitrage conversion (Table 12), we find that the two approaches produce very similar results.

Since banks' internal credit rating systems typically follow either log-normal or log-logistic distributions, their ratings can be converted across different banks - or mapped to rating agency scales - using the No-Regulatory-Arbitrage conversion method. This can be implemented through the relationships established in Proposition 4.5.2, Proposition 5.2.2, and Proposition 4.3.2.

5.3 Conversion from First-Hitting Time Default Model (PIT) Ratings into Agency Ratings

We apply Proposition 5.2.1. to convert the First-Hitting-Time credit ratings TiC_{FH} (see equation (12) in Proposition 4.4.1) into ratings comparable with those of S&P Global Ratings. This procedure also provides a conversion of Expected Default Frequency (EDF) and Distance-to-Default (DD) into the Standard & Poor's rating scale without introducing regulatory arbitrage.

Assume our First-Hitting **<u>CCM = 1.5</u>**. When we take different μ values, we obtain different First-Hitting credit ratings RS_{FH} and their corresponding default probabilities PD_{FH} using the following formulas (see Proposition 4.4.1).

$$TiC_{FH} = \frac{CCM}{\mu} = \frac{\sigma_A^2}{\ln^2\left(A_0 / D\right)}$$

$$PD_{FH} = \Phi\left(\sqrt{\frac{1}{CCM}}\left(\frac{1}{\sqrt{\mu}} - \sqrt{\mu}\right)\right) + e^{\frac{2}{CCM}}\Phi\left(-\sqrt{\frac{1}{CCM}}\left(\frac{1}{\sqrt{\mu}} + \sqrt{\mu}\right)\right)$$

First-Hitting credit rating has the following confidence level formula (see Proposition 4.5.2) with CML = $e^{1.35}$.

$$\alpha = CL_{FH}\left(CCM\right) = \Phi\left(\frac{\sqrt{e^{1.35}}}{CCM} - \frac{1}{\sqrt{e^{1.35}}}\right) + e^{\frac{2}{CCM}}\Phi\left(-\frac{\sqrt{e^{1.35}}}{CCM} - \frac{1}{\sqrt{e^{1.35}}}\right)$$

For **<u>CCM=1.5</u>**, we calculate **<u>α= 91.906%</u>**.

Standard & Poor's rating has Q = 0.625913 (see Proposition 4.2), its confidence level and rating formulae are as follows (see Proposition 4.5.2).

$$\alpha = CL_{SP}\left(CCM\right) = \Phi\left(\frac{1.35 - \frac{1}{0.625913}\cdot\ln CCM + \ln\left(CCM+1\right)/2}{\sqrt{\ln\left(CCM+1\right)}}\right)$$

$$\ln\left(TiC_{SP}\right) = 0.625913\cdot\Phi^{-1}\left(PD\right)\cdot\sqrt{\ln\left(CCM+1\right)} - \frac{0.625913}{2}\ln\left(CCM+1\right) + \ln(CCM)$$

We solve the equation **<u>91.906% = CL_{SP}（CCM*）</u>** to obtain **<u>CCM*=1.35373</u>**.

Table 13 lists the rating conversions corresponding to different μ values.

CCM =	μ	RiskScore	PD$_{FH}$	DD	EDF	S&P	S&P TTC	S&P
1.5	1	150	69.40%	0	50.00%	139	23.7%	CCC/C
α =	5	30	12.60%	1.46	7.21%	53.35	4.7%	B
91.91%	10	15	1.90%	2.32	1.01%	30.99	1.9%	BB-
	15	10	0.30%	2.95	0.16%	21.08	0.9%	BB
CCM* =	20	7.5	0.10%	3.47	0.03%	15.41	0.5%	BB+
1.35373	25	6	0.00%	3.92	0.00%	11.75	0.3%	BBB-
	30	5	0.00%	4.32	0.00%	9.23	0.2%	BBB
	35	4.3	0.00%	4.69	0.00%	7.41	0.1%	BBB+
	50	3	0.00%	5.66	0.00%	4.18	0.0%	A+
	75	2	0.00%	6.98	0.00%	1.93	0.0%	AAA
	100	1.5	0.00%	8.08	0.00%	1.01	0.1%	AAA
	150	1	0.00%	9.93	0.00%	0.34	0.0%	AAA

Table 13 Conversions of TiC$_{FH}$, EDF, and DD to S&P ratings when CCM=1.5

Similarly, for **CCM=5**, we have **72.749% = CL_{SP}(CCM*)** and get **CCM*=2.22928**, as shown in the following conversion table 14.

CCM =	μ	RiskScore	PD$_{FH}$	DD	EDF	S&P	S&P TTC	S&P
5	1	500	77.70%	0	50.00%	258.78	100.0%	C/D
α =	5	100	38.40%	0.8	21.19%	126.37	18.8%	CCC/C
72.75%	10	50	19.10%	1.27	10.15%	85.4	9.7%	CCC+
	15	33.3	10.10%	1.62	5.30%	65.12	6.4%	B-
CCM* =	20	25	5.50%	1.9	2.87%	52.41	4.6%	B
2.22928	25	20	3.10%	2.15	1.59%	43.55	3.4%	B+
	30	16.7	1.70%	2.37	0.89%	36.97	2.6%	BB-
	35	14.3	1.00%	2.57	0.51%	31.87	2.0%	BB-
	50	10	0.20%	3.1	0.10%	21.75	1.0%	BB
	75	6.7	0.00%	3.82	0.01%	13.02	0.3%	BBB-
	100	5	0.00%	4.43	0.00%	8.51	0.1%	BBB+
	150	3.33	0.00%	5.44	0.00%	4.21	0.0%	AA-

Table 14 Conversions of TiC$_{FH}$, EDF, and DD to S&P ratings when CCM=5

We can use Table 9 (Proposition 5.1) to derive the corresponding Standard & Poor's letter rating.

It should be noted that common EDF rating conversion methods simply convert EDF values into an institution's letter ratings, without considering the differences in μ values. This causes instable conversions for many banks.

We can view the difference between S&P TTC and PD_{FH} (or EDF) as an outlook on the future. as PD_{FH}/EDF is the Point-In-Time default probability. Their difference is a measure for the difference between short-term risk and long-term risk. If we believe that ultimately, risk will revert to the Through-The-Cycle default probability, then this difference reflects a future trend. The gap between the two therefore captures the divergence between short-term and long-term credit risk. If one assumes that credit risk eventually reverts toward the TTC level, this difference can be interpreted as an indicator of the future trajectory of credit risk.

PROPOSITION 5.3 (Credit Risk Outlook) *Let us define*

$$Outlook = PD_{FH} - \text{S\&P } TTC \tag{28}$$

If Outlook>0, then the future trend is positive; if Outlook<0, then the trend is negative; if Outlook =0, then the trend is neutral.

6 Summary

Time-Consistent (TiC) credit rating uses a **Behavior-specific** approach, involving the distribution of default time (τ) and the default probability. It consists of two essential risk drivers. Other rating approaches can be considered special cases of the Behavior-specific approach.

Rating agencies use a **Process-specific** approach, not knowing the theoretical distribution of the default time (τ), nor the formula for calculating the default probability. We estimate the distribution of τ and the default probability from observed data.

Bank ratings use a **Distribution-specific** approach. Although the distribution of τ is unknown, it is assumed to be either a Log-normal or Log-logistic distribution, and a statistical default probability formula is derived using historical data modeling.

Academic credit ratings use **Default-specific** approach, deriving both the theoretical distribution of τ and the theoretical default probability formula.

REFERENCES

Avellaneda, Marco and Zhu, Jingyi. (2001) *Distance to default*, *www.risk.net*, 2001.

Basel Committee on Banking Supervision (Basel). (2006) Basel II: International Convergence of Capital Measurement and Capital Standards: A Revised Framework -

Comprehensive Version, *Basel Framework:* *https://www.bis.org/basel_framework/* June 2006.

Cantor, R. and Falkenstein E. (2001) Testing For Rating Consistency In Annual Default Rates, *The Journal of Fixed Income* September, 2001, 36–51.

Carlehed M and Petrov A. (2012) A methodology for point-in-time–through-the-cycle probability of default decomposition in risk classification systems, *Journal of Risk Model Validation* Volume 6/Number 3, Fall 2012 (3–25).

Crosbie, P., and Bohn, J. (2002) Modeling Default Risk, *KMV*. Moody's KMV.

Financial Accounting Standards Board (FASB), (2016), Financial Instruments—Credit Losses
(Topic 326), *Financial Accounting Series: Accounting Standards Updates* No. 2016-13, June 2016.

Hamilton, D. (2011) An Introduction to Through-the-Cycle Public Firm EDF™ Credit Measures, *Quantitative Credit Research* May, 2011.

International Accounting Standards Board (IASB), (2014), International Financial Reporting Standard 9 (IFRS 9) - Financial Instruments, *Basis for Conclusions International Financial Reporting Standard* July 2014.

Liang, J., Wu, Y. and Hu, B. (2016) Asymptotic traveling wave solution for a credit rating migration problem, *Journal of Differential Equations*, 261, 1017–1045.

Moody's Investors Service (2011), (1920-2010) Corporate Default and Recovery Rates, *Special Comment*. Moody's Investors Service

S&P Global Ratings. (2017) Annual Global Corporate Default Study And Rating Transitions, *RatingsDirect*. S&P Global Ratings

US SEC. (2012) Report to Congress Credit Rating Standardization Study, *U.S. Securities and Exchange Commission* September, 2012. https://www.sec.gov/files/939h_credit_rating_standardization.pdf

Valuˇzis, M. (2008) On the Probabilities of Correlated Defaults: a First Passage Time Approach, *Nonlinear Analysis: Modelling and Control* Vol. 13, No. 1, 117–133.

Yang, Yimin. (2021) Time-Consistent Credit Risk Rating – Approach and Applications, *World Finance Conference,* Norway, August, 2021.

Yang, Yimin. (2019) Theory and applications of Time-Consistent Credit Rating, *Fifth Annual Volatility Institute Conference at New York University: Credit Risk in a New Era,* Shanghai, China, November 22, 2019.